Interview Transcript with Dr. Kavita Joshi (Pune, India)   Senior Epidemiologist, ICMR-NIE

Interviewer: Dr. Joshi, thank you for joining us. To begin, could you tell us about your background and your role during the pandemic? Dr. Joshi: Thank you. I m Dr. Kavita Joshi, a public health epidemiologist with the Indian Council of Medical Research (ICMR), currently posted at the National Institute of Epidemiology in Pune. My work primarily involves infectious disease surveillance, outbreak modeling, and public health policy advisory.

When COVID-19 began, my team was tasked with two major responsibilities: first, designing and refining epidemiological models for India s response strategy; and second, coordinating state-level data integration and field surveillance. It was the largest and most complex public health challenge we d ever faced.

Interviewer: What were those early days like, when the first cases began to appear? Dr. Joshi: In January 2020, we were monitoring the outbreak in Wuhan closely. By late February, it became clear that India wouldn t remain untouched. The first confirmed cases in Kerala were our wake-up call. Within ICMR, we began developing a national testing protocol and contact tracing framework.

Back then, the biggest challenge wasn t the virus itself   it was data uncertainty. We were working with fragmented reports, limited testing, and incomplete transmission chains. Epidemiology thrives on data, but in those first months, we were often working in the dark.

Interviewer: How did India s testing capacity evolve over time? Dr. Joshi: Tremendously. In March 2020, we had barely 10 authorized testing labs in the entire country. By July, we had over 1,000. That expansion   both in RT-PCR and later antigen testing   was unprecedented.

We created a hub-and-spoke model: ICMR s central labs served as hubs that trained and validated regional labs. States like Maharashtra, Tamil Nadu, and Karnataka rapidly scaled up testing due to strong biomedical networks. But many northern and northeastern states lagged initially due to logistics and infrastructure.

Testing is not just about machines   it s about reagents, data systems, and trained personnel. We had to think like logisticians, not just scientists.

Interviewer: How did the team balance epidemiological advice with political and social realities? Dr. Joshi: That s the tightrope every public health scientist walks. Epidemiology tells you what should be done; governance decides what can be done. During lockdown decisions, for example, we modeled multiple scenarios   from total shutdowns to regional containment   and presented likely outcomes.

Sometimes, the science was followed to the letter. Other times, socio-economic constraints took precedence. We learned to frame evidence not as orders but as options with trade-offs. I respect policymakers for the difficult choices they made, even when they diverged from ideal epidemiological recommendations.

Interviewer: How did the Delta wave change your understanding of the virus s behavior? Dr. Joshi: The Delta variant was an epidemiologist s nightmare. It spread faster, caused higher viral loads, and shifted infection patterns toward younger populations. The R0 (basic reproduction number) doubled in weeks. Our earlier models had to be recalibrated daily.

What was most striking was how heterogeneous the epidemic became   urban centers peaked and declined while rural districts were still climbing. It wasn t a single curve; it was a hundred overlapping curves. That taught us that national averages can hide local crises.

Interviewer: How did rural transmission affect India s overall response? Dr. Joshi: Profoundly. Rural India was both resilient and vulnerable. Resilient because of lower population density and open-air lifestyles, but vulnerable due to healthcare access.

We saw that once the virus reached villages, mortality was driven not by viral severity but by delayed diagnosis and lack of oxygen. Our surveillance teams had to pivot   instead of large cities, we deployed mobile testing vans to rural clusters.

Data gaps were huge. Many deaths went unregistered. That was painful as an epidemiologist because behind every missing number was a human story. We later adjusted estimates using seroprevalence surveys, which revealed that true infection rates were far higher than reported.

Interviewer: Could you describe how the national serosurveys were conducted? Dr. Joshi: Yes. The ICMR serosurveys were among our most important tools. We conducted four major rounds between 2020 and 2021, sampling tens of thousands across urban and rural areas. The goal was to estimate how many people had antibodies   essentially, how far the virus had already spread.

Each survey required enormous coordination: selecting random households, obtaining informed consent, transporting samples under cold chain, and analyzing them in reference labs. By mid-2021, the fourth serosurvey showed that over 67% of Indians had antibodies   indicating far wider exposure than case numbers suggested.

That data shaped the vaccination strategy   we realized population-level immunity was rising, but unevenly distributed.

Interviewer: What were some key lessons in communication and public trust during this crisis? Dr. Joshi: Transparency is everything. The public will forgive uncertainty but not inconsistency. When guidance changes   and it will in any pandemic   the why must be explained. Early on, when we changed mask policies, the lack of context created confusion.

We learned to communicate uncertainty as part of science. It s okay to say, We know this much, but we re still learning. The other lesson was the need for one consistent public voice. Multiple experts giving conflicting opinions on TV hurt credibility. Public health communication must be unified, empathetic, and data-backed.

Interviewer: Were there moments of professional or personal exhaustion for you and your team?

Dr. Joshi: Many. We worked 18-hour days for months, sometimes sleeping in offices. The emotional fatigue of analyzing death data daily was immense. Many of us caught COVID ourselves. I lost two colleagues.

What kept us going was purpose. Every dataset, every line of code, every field report felt like part of something larger. In public health, you rarely get instant gratification but during COVID, every decision could save thousands. That sense of urgency gave meaning to exhaustion.

Interviewer: What innovations or data tools emerged that you think should continue beyond the pandemic? Dr. Joshi: Several. 1. The ICMR COVID Data Portal integrating lab results nationwide was a game-changer. For the first time, we had near real-time epidemiological data at a national scale. 2. The CoWIN platform proved that India can manage massive digital public health systems. 3. At the state level, data dashboards helped local officials plan oxygen, beds, and staff dynamically.

These tools shouldn t vanish post-pandemic. They can transform tuberculosis surveillance, maternal health tracking, and future outbreak response.

Interviewer: India s vaccination drive was among the world s largest. What stood out to you about that process? Dr. Joshi: The scale was breathtaking over a billion doses in under a year. We built not just logistics but public confidence. Initially, vaccine hesitancy was significant, especially in rural areas. Field teams used community influencers teachers, priests, self-help groups to spread awareness.

We also relied heavily on local languages and visual aids rather than English press releases. The moment we localized the message, uptake soared. By late 2021, most districts had crossed 80% coverage for adults. It was one of the rare times when science, administration, and community aligned perfectly.

Interviewer: What do you believe are India s biggest takeaways from the pandemic? Dr. Joshi: Three main takeaways: 1. Data is infrastructure. Without accurate data, decisions are guesswork. 2. Public health must be decentralized. District-level autonomy saved lives. 3. Preparedness needs permanence. We shouldn t dismantle COVID infrastructure once the emergency fades it should evolve into an all-hazard response system.

We must also invest in field epidemiology training. India needs at least 10,000 skilled epidemiologists we have only a fraction of that. Pandemics will return, and we must be ready.

Interviewer: Personally, what has this journey taught you about leadership and science? Dr. Joshi: It taught me humility. Models and graphs are abstractions of real lives. Behind every curve is someone s father or mother. That perspective grounded us.

It also reaffirmed my faith in collaboration between disciplines, between states, between generations of scientists. During one serosurvey, a 22-year-old intern and a 60-year-old lab technician worked side by side in rural Bihar. That s what science looks like in action collective effort across hierarchies.

And finally, it taught me that science and empathy must travel together. Data may guide the mind, but compassion sustains the soul.

Interviewer: That s beautifully said, Dr. Joshi. Thank you for your insight and service.   Dr. Joshi: Thank you. The pandemic challenged us, but it also reminded us why public health exists to serve, protect, and learn from every life.

--- End of Interview ---